

Lecture 2 - Crystal Geometry

R. Hielscher

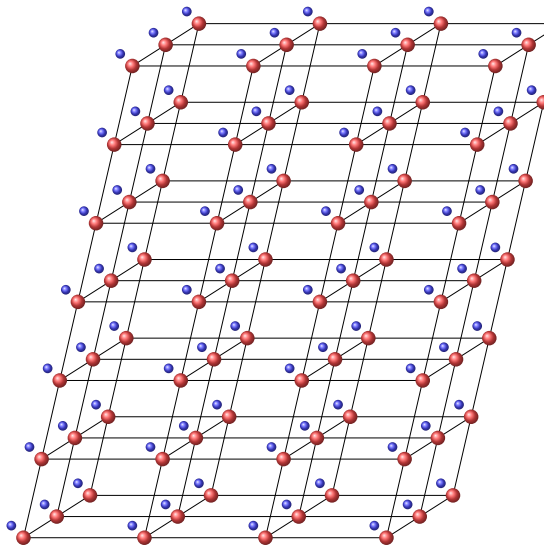
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MTEX Workshop 2024

Crystals

Definition

A crystal is an anisotropic, homogenous body consisting of a three-dimensional periodic ordering of atoms, ions or molecules.



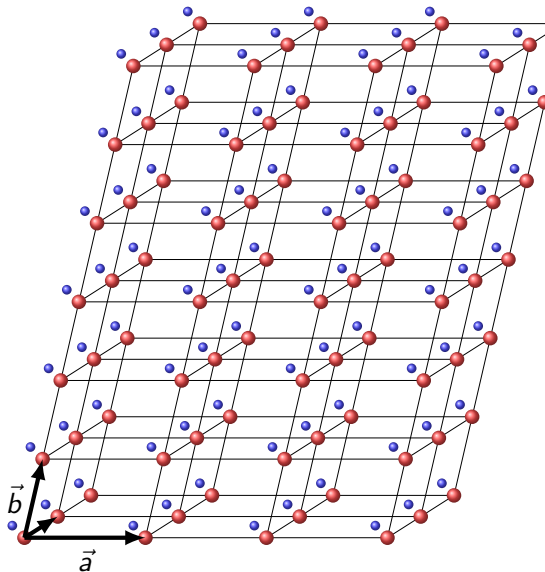
Crystals

Definition

A crystal is an anisotropic, homogenous body consisting of a three-dimensional periodic ordering of atoms, ions or molecules.

Consequence: There are linearly independent vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ such that the atomic structure is invariant with respect to translations about these vectors.

- ▶ The definition of periodicity assumes infinity of the atomic structure.
- ▶ The choice of the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ is not unique



The Unit Cell

- ▶ is the parallelepiped spanned by $\vec{a}$, $\vec{b}$, $\vec{c}$.
- ▶ is the smallest region that constitutes a repeating pattern.
- ▶ is characterized by the length and angles

$$a = |\vec{a}|, b = |\vec{b}|, c = |\vec{c}|$$

$$\alpha = \angle(\vec{b}, \vec{c}), \beta = \angle(\vec{a}, \vec{c}), \gamma = \angle(\vec{b}, \vec{a}).$$

```
abc = [5.2, 5.2, 5.3]
```

```
abg = [90 99.25 90] * degree;
```

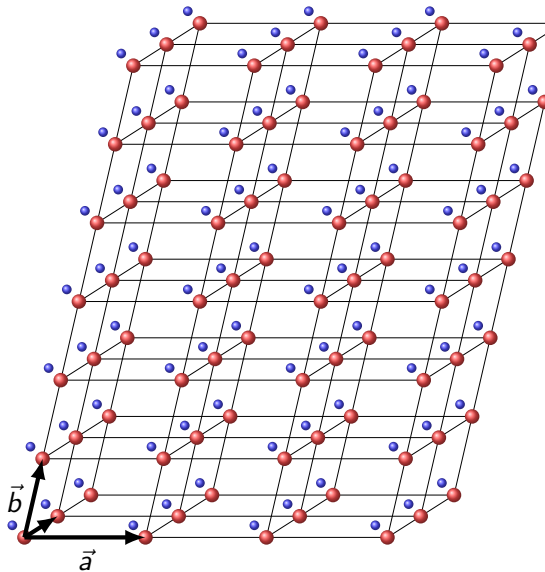
```
cs = crystalSymmetry( '1', abc, abg)
```

```
ans = crystalSymmetry
```

```
symmetry          : 1
```

```
a, b, c           : 5.2, 5.2, 5.3
```

```
alpha, beta, gamma: 90, 99.25, 90
```



The Crystal Coordinate Systems

The vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ constitute a right handed but non orthogonal coordinate system.

Lattice points are given by coordinates u, v, w :

$$\cdot uvw \cdot = \vec{d} = u\vec{a} + v\vec{b} + w\vec{c}.$$

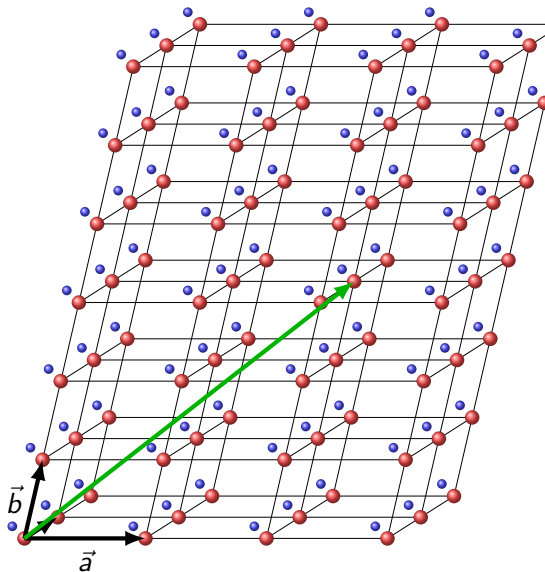
Usage: slip systems, dislocation systems

`d = Miller (1,1,0,cs, 'uvw')`

`d = Miller (1)`

u	v	w
1	1	0

Smaller integers u, v, w indicate higher density of lattice points on the line and larger distance between parallel lines.



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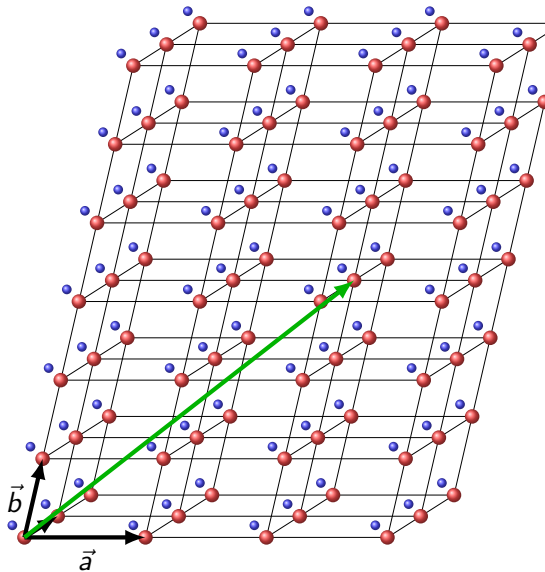
`d = Miller (1,1,0,cs, 'uvw')`

Lattice directions $[uvw]$: all vectors parallel to $\vec{d}$

`d == [2*d, -d]`

ans =

1 0



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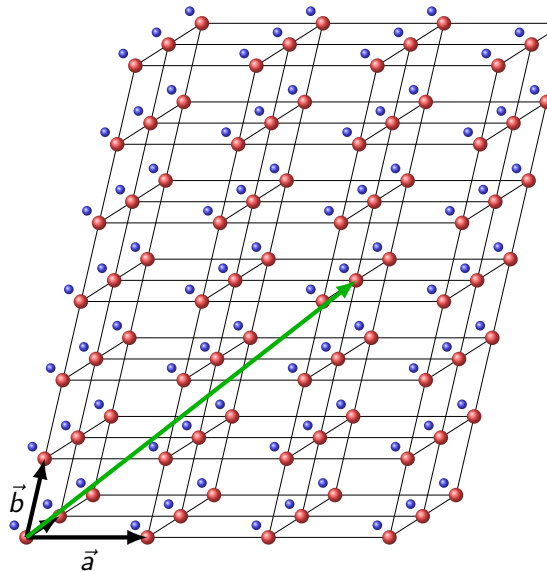
Lattice directions $[uvw]$: all vectors parallel to $\vec{d}$

```
d == [2*d, -d]
```

```
[abs(d), norm(2*d)]
```

```
ans =
```

```
1.4142    2.8284
```



The Crystal Coordinate Systems

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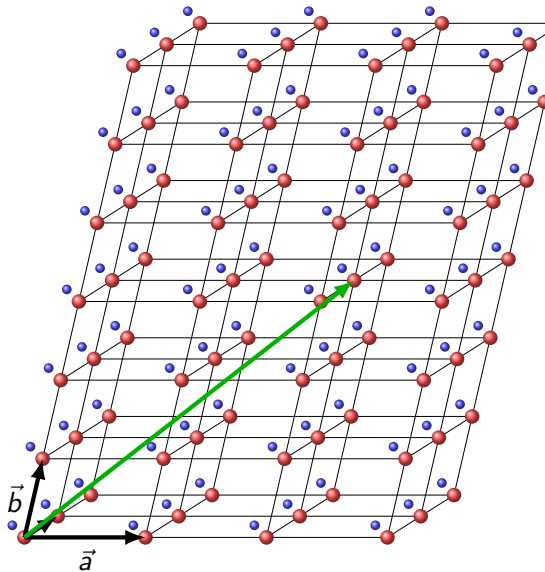
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Lattice directions $[uvw]$: all vectors parallel to $\vec{d}$

`d == [2*d, -d]`

`[abs(d), norm(2*d)]`

Examples: $[100]$, $[1\bar{1}0] = [2\bar{2}0] \neq [\bar{1}10]$



The Reciprocal Coordinate System

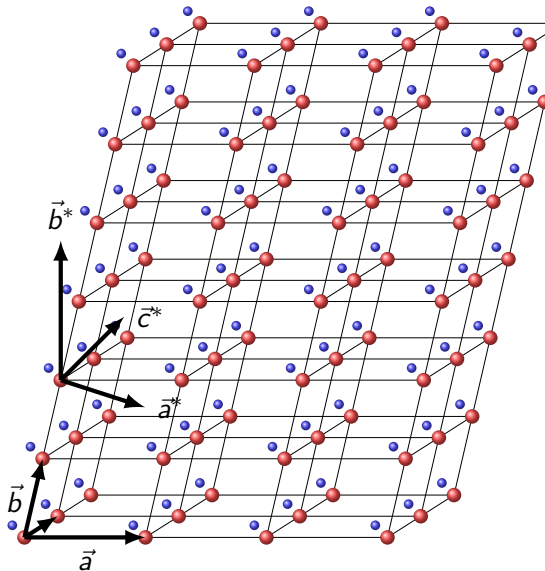
The reciprocal axes are orthogonal to $\vec{a}, \vec{b}, \vec{c}$,

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V}, \quad \vec{b}^* = \frac{\vec{c} \times \vec{a}}{V}, \quad \vec{c}^* = \frac{\vec{a} \times \vec{b}}{V}$$

- ▶ $V = \vec{a} \cdot (\vec{b} \times \vec{c})$ is the volume of the unit cell
- ▶ $\vec{a} \cdot \vec{a}^* = 1, \vec{b} \cdot \vec{b}^* = 1, \vec{c} \cdot \vec{c}^* = 1$
- ▶ units of $\vec{a}^*, \vec{b}^*, \vec{c}^*$ are reciprocal to $\vec{a}, \vec{b}, \vec{c}$
- ▶ for cubic lattices $\vec{a} \parallel \vec{a}^*, \vec{b} \parallel \vec{b}^*, \vec{c} \parallel \vec{c}^*$,

Directions with respect to the $\vec{a}^*, \vec{b}^*, \vec{c}^*$ are given by integers coordinates h, k, ℓ

$$\vec{n} = h\vec{a}^* + k\vec{b}^* + \ell\vec{c}^*$$

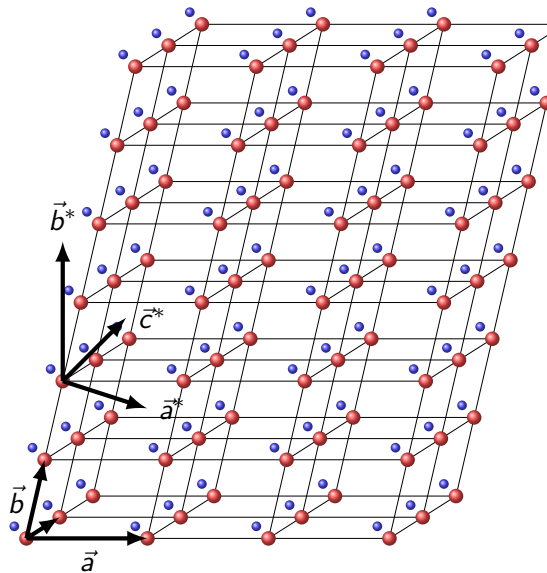


The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs,'hkl')
```

```
d = Miller(1)
```

h	k	l
1	1	0



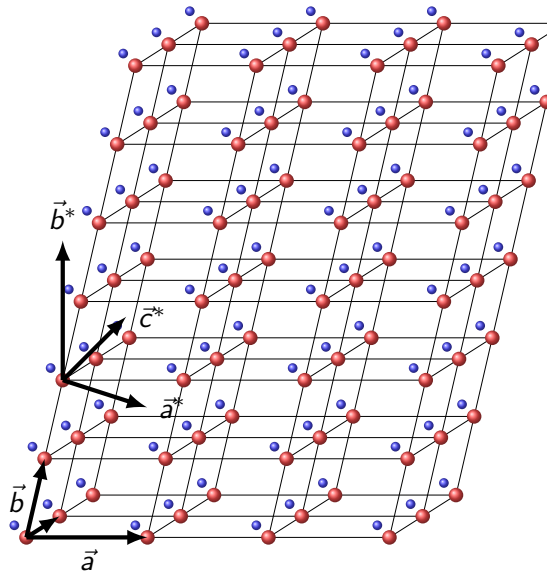
The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs,'hkl')
```

```
Miller({1,1,0},{0,1,1},cs,'hkl')
```

```
ans = Miller (1)
```

h	k	l
1	1	0

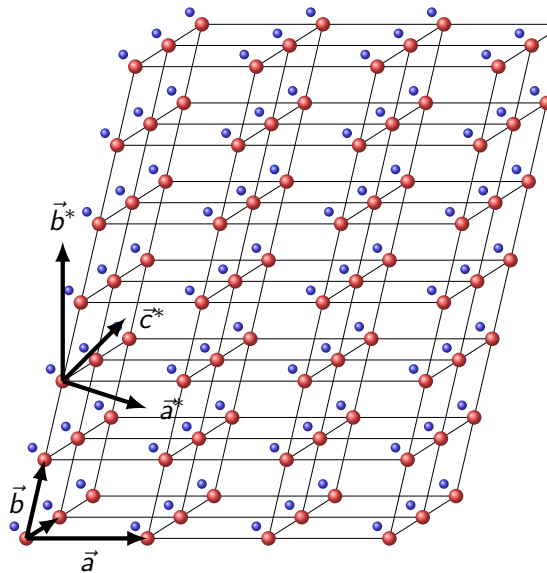


The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs,'hkl')
```

```
Miller({1,1,0},{0,1,1},cs,'hkl')
```

```
rec = [cs.aAxisRec, cs.bAxisRec, ...  
       cs.cAxisRec]
```



The Reciprocal Coordinate System

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```

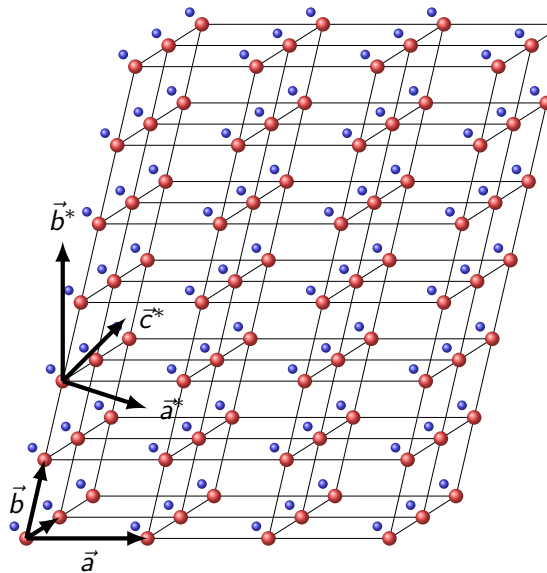
```
Miller({1,1,0},{0,1,1},cs, 'hkl')
```

```
rec = [cs.aAxisRec, cs.bAxisRec, ...  
       cs.cAxisRec]
```

```
dot(rec, cs.aAxis)
```

```
ans =
```

```
1      0      0
```



The Reciprocal Coordinate System

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n = Miller(1,1,0,cs,'hkl')
```

```
Miller({1,1,0},{0,1,1},cs,'hkl')
```

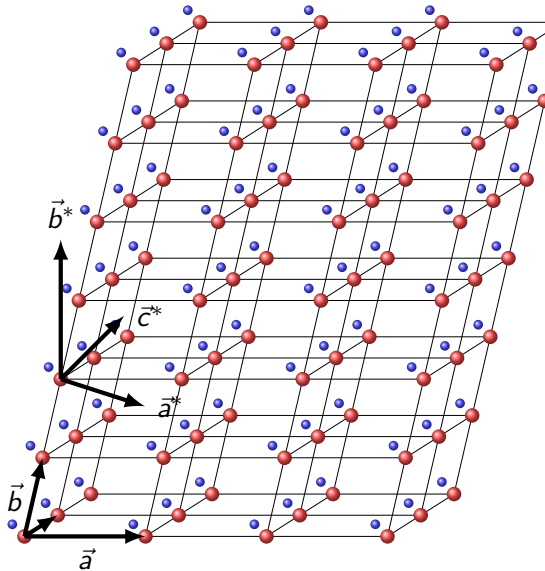
```
rec = [cs.aAxisRec, cs.bAxisRec, ...  
       cs.cAxisRec]
```

```
dot(rec, cs.aAxis)
```

```
cross(cs.aAxisRec, cs.bAxisRec)
```

```
ans = Miller (1)
```

u	v	w
0	0	1



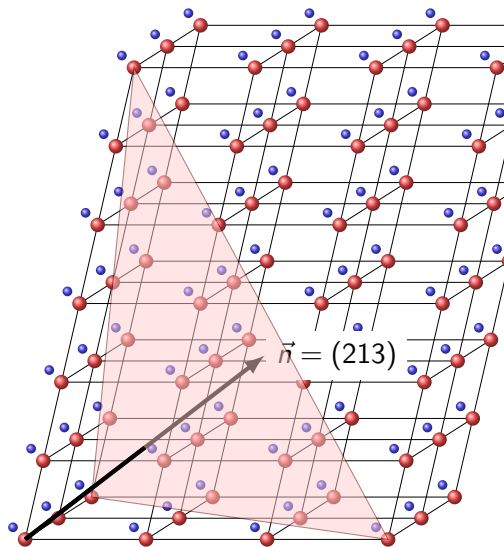
Crystal Lattice Planes

The plane P normal to $\vec{n} = h\vec{a}^* + k\vec{b}^* + \ell\vec{c}^*$ is

$$P = \{\vec{x} \mid \vec{x} \cdot \vec{n} = d\} = (hkl).$$

It intersects the $\vec{a}$ -axis in $\cdot u00 \cdot$ if

$$d = u\vec{a} \cdot \vec{n} = u\vec{a} \cdot (h\vec{a}^* + k\vec{b}^* + \ell\vec{c}^*) = uh.$$



Crystal Lattice Planes

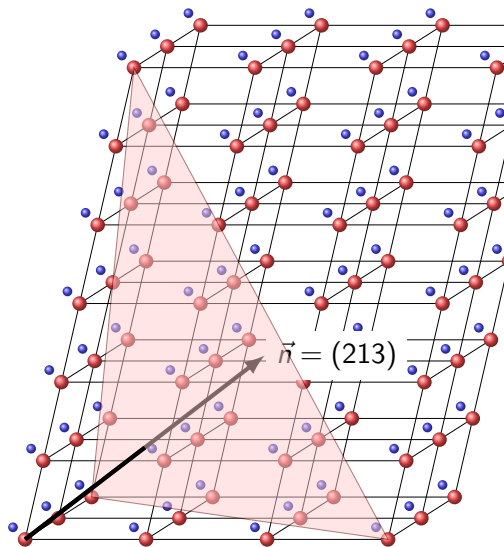
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Hence, (hkl) intersects $\vec{a}$, $\vec{b}$ and $\vec{c}$ in $\frac{d}{h}$, $\frac{d}{k}$, $\frac{d}{\ell}$.



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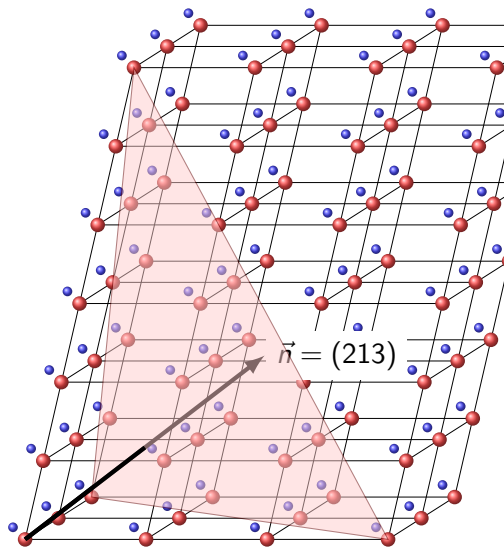
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Hence, (hkl) intersects $\vec{a}$, $\vec{b}$ and $\vec{c}$ in $\frac{d}{h}$, $\frac{d}{k}$, $\frac{d}{\ell}$.

Given axes intersections $m, n, p \in \mathbb{N}$. The corresponding normal vector is

$$\vec{n} = \frac{d}{m}\vec{a}^* + \frac{d}{n}\vec{b}^* + \frac{d}{p}\vec{c}^*$$

with $d = \text{LCM}(m, n, p)$.



Crystal Lattice Planes

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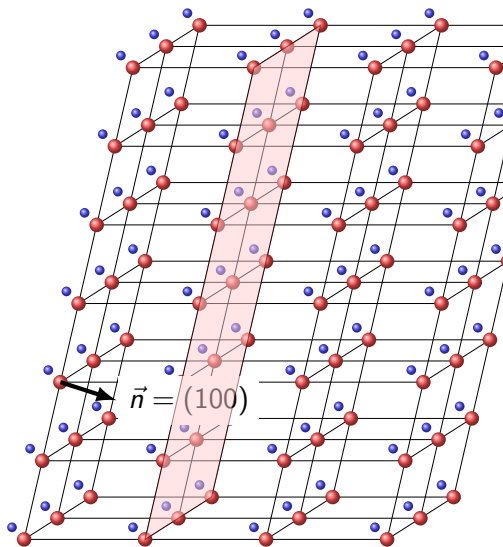
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$$k = 0 \iff \frac{1}{0} = \infty \iff (hkl) \text{ is parallel to } \vec{b}$$



Crystal Lattice Planes

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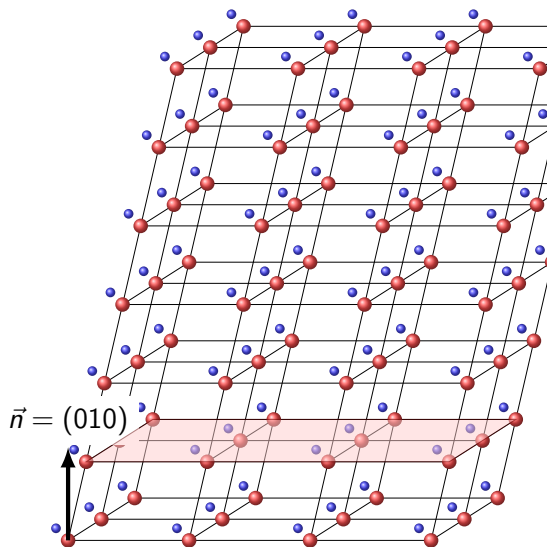
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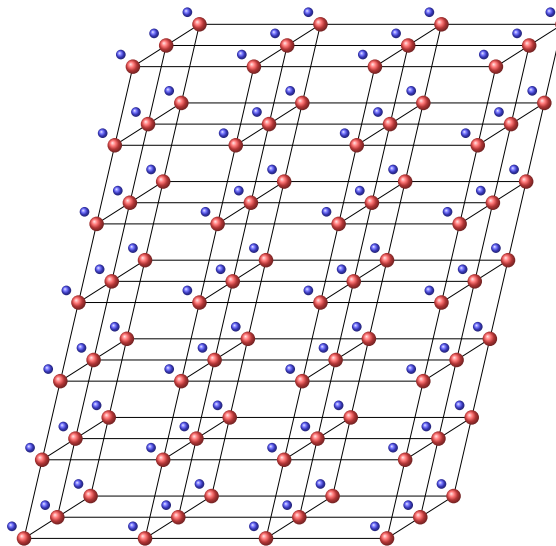
$k = 0 \iff \frac{1}{0} = \infty \iff (hkl)$ is parallel to $\vec{b}$



Miller Indices

The lattice plane (123) perpendicular to $\vec{n} = \vec{a}^* + 2\vec{b}^* + 3\vec{c}^*$ which intersects the direct lattice axes $\vec{a}$, $\vec{b}$ and $\vec{c}$ at 1, $\frac{1}{2}$ and $\frac{1}{3}$.

- ▶ The coordinates h, k, ℓ are the **Miller indices** of the plane P .
- ▶ planes (hkl) parallel to $\vec{a}$ have interception point $\frac{1}{\infty}$ and hence $h = 0$
- ▶ planes $(h00)$ perpendicular to $\vec{a}^*$ have interception points $(\frac{d}{h}, \frac{1}{\infty}, \frac{1}{\infty})$



The Zone Equation

A direction $[uvw]$ is parallel to a plane (hkl) if

$$hu + kv + \ell w = 0.$$

```
d1 = Miller(1,-1,2,cs,'uvw')  
n1 = Miller(1,1,0,cs,'hkl')  
dot(d1,n1)
```

```
ans =
```

```
7.7716e-16
```

The Zone Equation

A direction $[uvw]$ is parallel to a plane (hkl) if

$$hu + kv + \ell w = 0.$$

```
d1 = Miller(1,-1,2,cs,'uvw')  
n1 = Miller(1,1,0,cs,'hkl')  
dot(d1,n1)
```

```
angle(n1,d2) ./ degree
```

```
ans =
```

```
90.0000
```

The Zone Equation

A direction $[uvw]$ is parallel to a plane (hkl) if

$$hu + kv + \ell w = 0.$$

The plane (hkl) spanned by the directions $[u_1 v_1 w_1]$ and $[u_2 v_2 w_2]$:

$$(hkl) = [u_1 v_1 w_1] \times [u_2 v_2 w_2]$$

```
d1 = Miller(1,-1,2,cs,'uvw')  
n1 = Miller(1,1,0,cs,'hkl')  
dot(d1,n1)
```

```
angle(n1,d2) ./ degree
```

```
d2 = Miller(1,0,0,cs,'uvw')  
cross(d1,d2)
```

```
ans = Miller (1)  
      h   k   l  
      0   2   1
```

The Zone Equation

A direction $[uvw]$ is parallel to a plane (hkl) if

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The plane (hkl) spanned by the directions $[u_1 v_1 w_1]$ and $[u_2 v_2 w_2]$:

$$(hkl) = [u_1 v_1 w_1] \times [u_2 v_2 w_2]$$

The common direction $[uvw]$ of two planes $(h_1 k_1 \ell_1)$ and $(h_2 k_2 \ell_2)$

$$[uvw] = (h_1 k_1 \ell_1) \times (h_2 k_2 \ell_2)$$

```
d1 = Miller(1,-1,2,cs,'uvw')
n1 = Miller(1,1,0,cs,'hkl')
dot(d1,n1)
```

```
angle(n1,d2) ./ degree
```

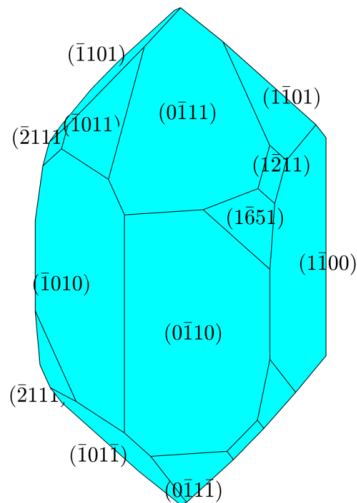
```
d2 = Miller(1,0,0,cs,'uvw')
cross(d1,d2)
```

```
n2 = Miller(1,0,0,cs,'hkl')
cross(n1,n2)
```

```
ans = Miller (1)
      u   v   w
      0   0  -1
```


Morphology

Crystals are formed by crystallization from a solution or a melt.
the two steps of crystallization: nucleation and growth

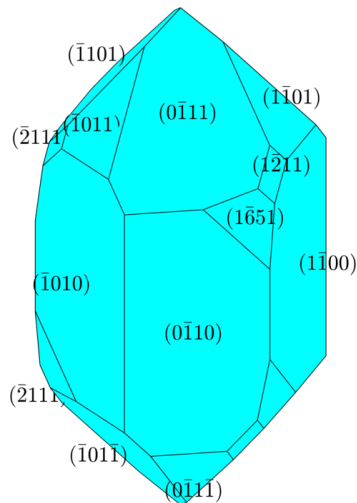


Morphology

Crystals are formed by crystallization from a solution or a melt.
the two steps of crystallization: nucleation and growth

- ▶ the grow rate is an anisotropic property
- ▶ lattice planes with high grow rate get smaller
- ▶ lattice planes with low grow rate become larger
- ▶ lattice planes (hkl) with small Miller indices usually have high density and grow slowly
- ▶ the shape of a crystal is formed by its low index lattice planes
- ▶ law of constant dihedral angles

```
cs = crystalSymmetry.load('quartz.cif')  
cS = crystalShape.quartz;  
plot(cS, 'colored')
```

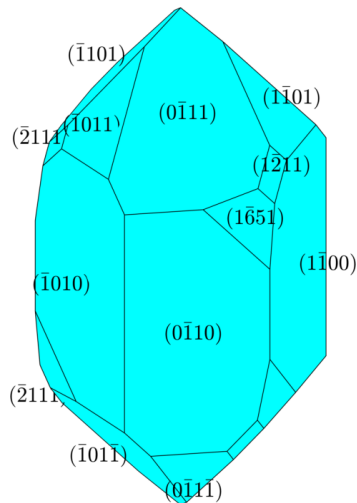


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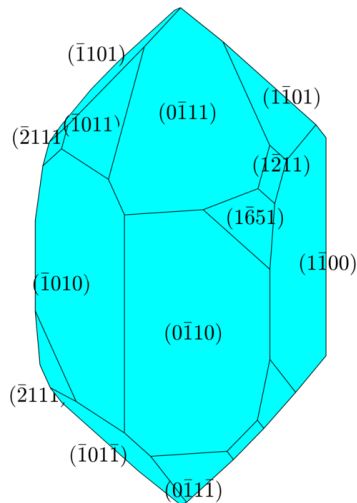


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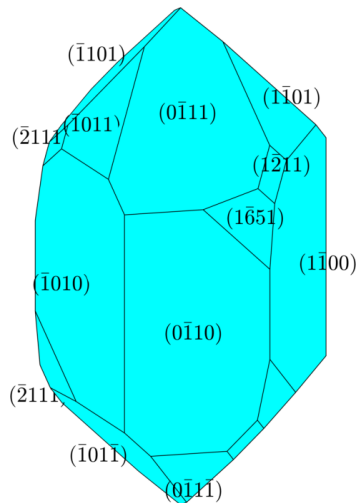


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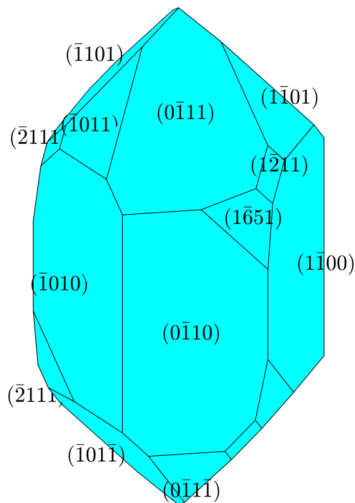


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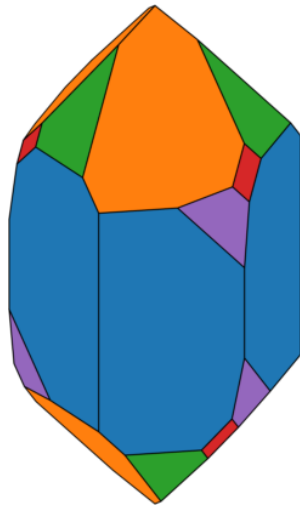


Morphology

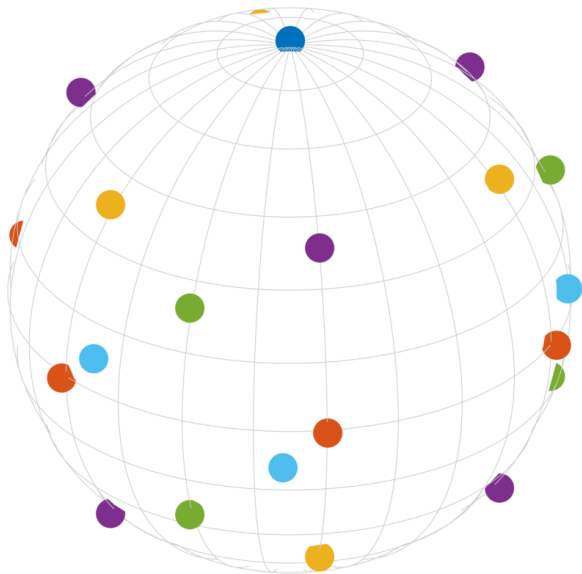
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```
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plot(cS, 'colored')
```



The Crystal Planes of Quartz in a Stereographic Projection



c c-axis:

(0001) , $(000\bar{1})$

m hexagonal prism:

$(10\bar{1}0)$, $(01\bar{1}0)$, $(1\bar{1}00)$

$(\bar{1}010)$, $(0\bar{1}10)$, $(\bar{1}100)$

r positive rhomboeder:

$(10\bar{1}1)$, $(\bar{1}101)$, $(0\bar{1}11)$

z negative rhomboeder:

$(1\bar{1}01)$, $(01\bar{1}1)$, $(\bar{1}011)$

s trigonal bipyramid:

$(11\bar{2}1)$, $(1\bar{2}11)$, $(\bar{2}111)$

x positive trapezohedron:

$(51\bar{6}1)$, $(\bar{6}511)$, $(1\bar{6}51)$